

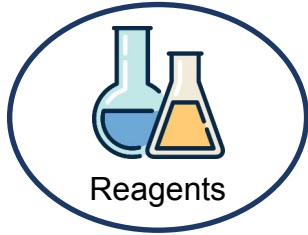
# Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

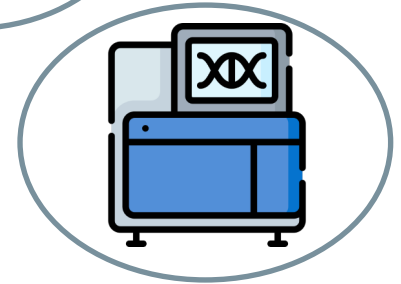
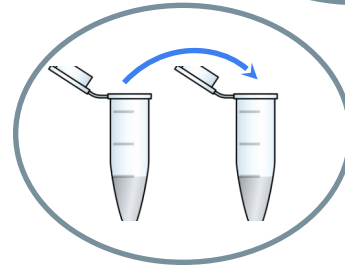
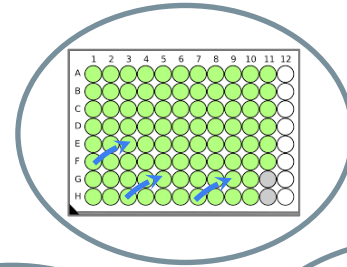


Background models

# Contamination will happen!



External  
Contamination

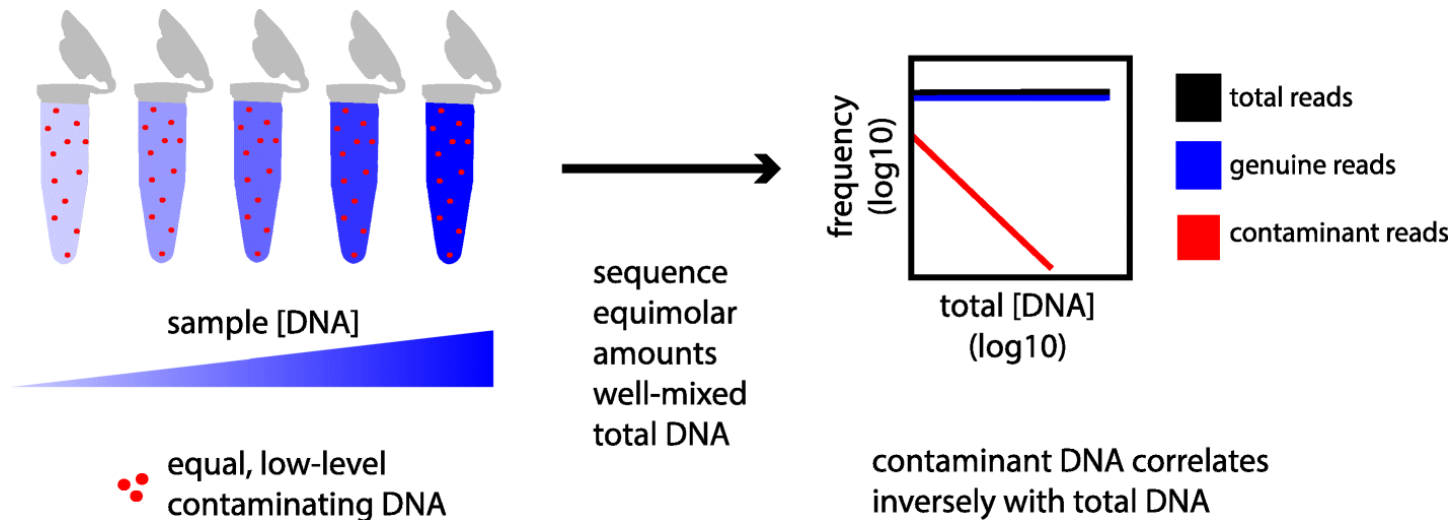


Internal  
Contamination  
(cross-contamination)

# Contamination can lead to inaccurate data interpretation

- Falsely inflates within-sample diversity
- Obscures differences between samples
- Interferes with comparisons across studies

# Contamination disproportionately affects low-biomass samples



# Water controls can help

- 'Blanks' such as water controls can help identify both internal and external contaminants
- Water controls should be processed the same way as the samples, using the the same reagents
  - Have at least 2 water controls, but the more the better!

# Background model

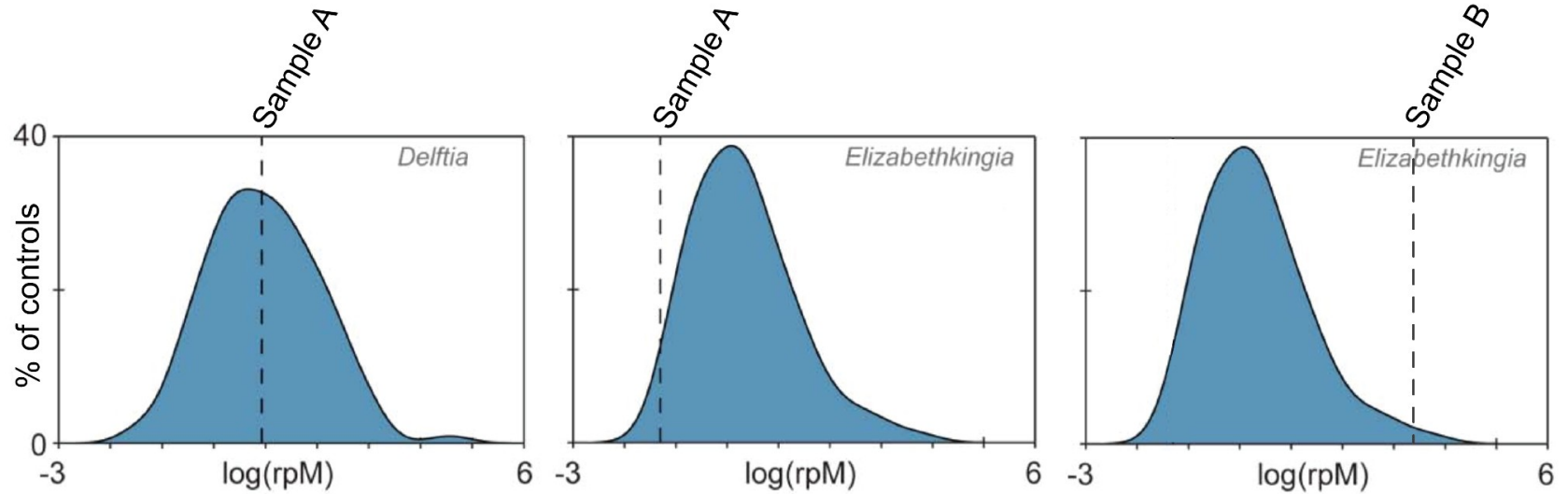
- Used to calculate relative abundance of the taxa found in negative controls
- 2 types of background models based on relative abundance metric:
  - **Standard:**
    - Uses the reads per million (rpm) as the metric of relative abundance.
  - **Mass normalized:**
    - Uses total ERCC reads to normalize taxon reads.
    - Only available when samples have been spiked with **ERCCs**.

# Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
  - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?

Background distribution  
overlaid with sample taxa abundance



--- Abundance in potentially infectious sample

 Distribution of abundance in controls

# Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
  - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?
- It is measured in terms of standard deviations from the mean

$$Z = \frac{x - \mu}{\sigma}$$

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

**Z-score linear scale:**



0

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$$Z = 0$$


$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

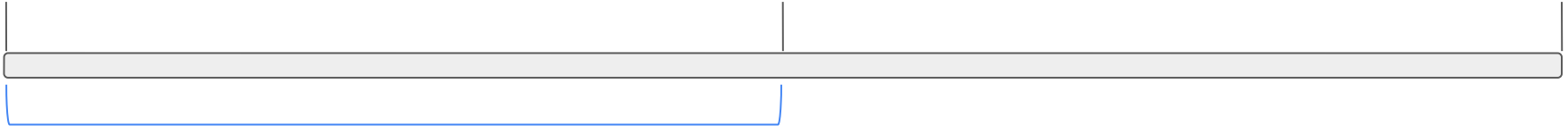
$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$Z < 0$

$Z = 0$



Taxa **more abundant in negative controls** than  
in samples

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

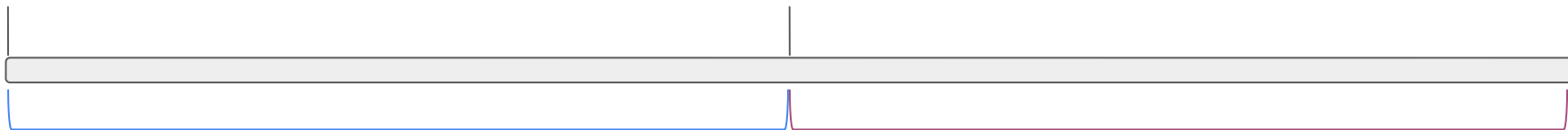
$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$Z < 0$

$Z = 0$

$Z > 0$



Taxa **more abundant in negative controls** than  
in samples

Taxa **more abundant in samples** than negative  
controls

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

Only in  
control  
**-100**

No difference

0

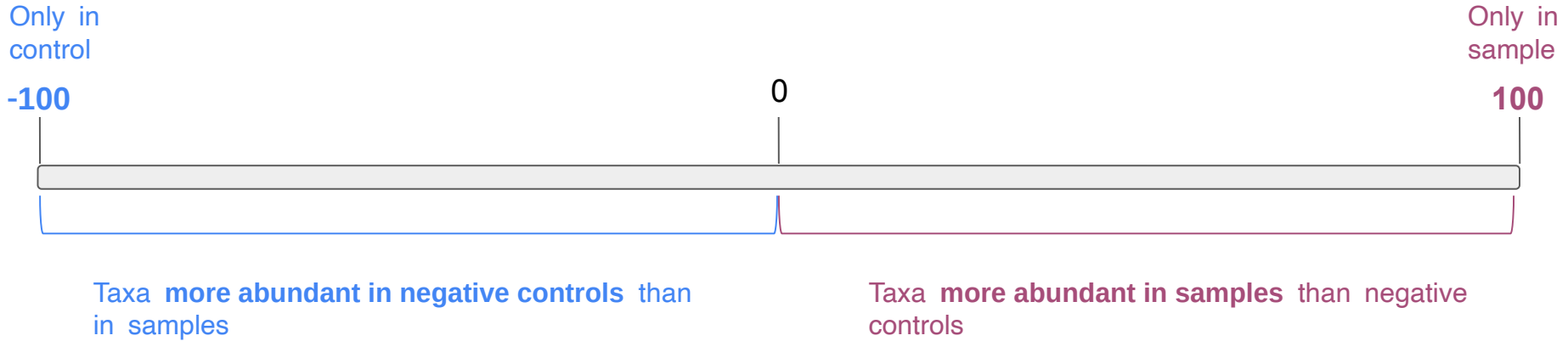
Only in  
sample  
**100**

Taxa **more abundant in negative controls** than  
in samples

Taxa **more abundant in samples** than negative  
controls

# Use Z-score filter after applying background model

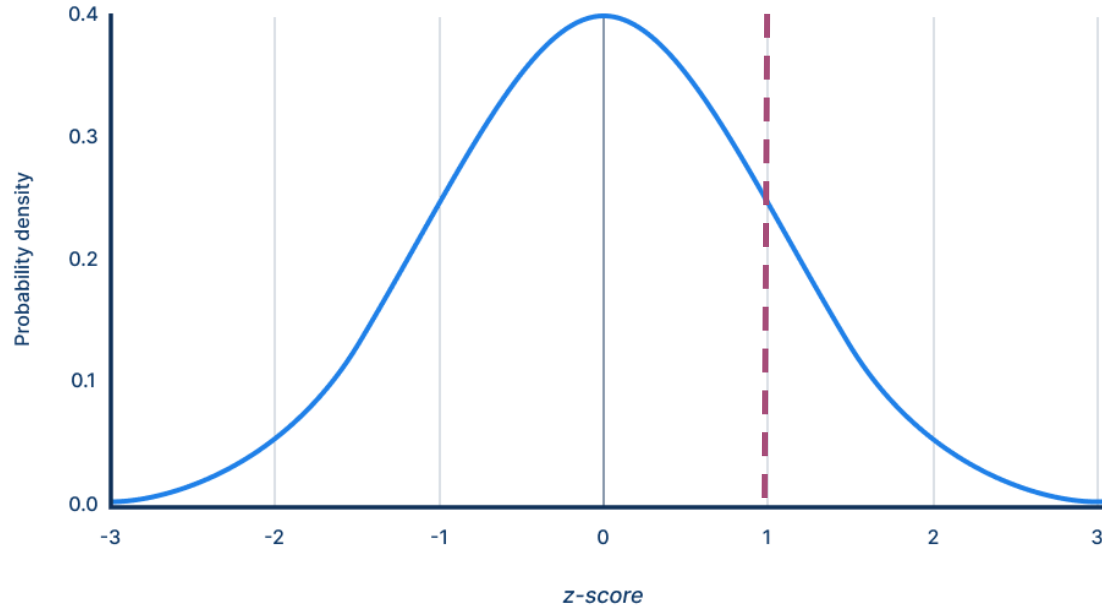
Start with taxa with Z-scores  $> 1$



# Use Z-score filter after applying background model

Start with taxa with Z-scores  $> 1$

**Standard normal distribution**



# Calculating Z-score

Mass normalized (mass\_norm) Z-score calculation:

$$Z = \frac{x - \mu}{\sigma} = \frac{(\text{mass\_norm of taxa } T \text{ in sample } S) - \text{mean}(\text{mass\_norm of taxa } T \text{ in background model samples})}{\text{std\_dev}(\text{mass\_norm of taxa in background model samples})}$$

Standard Z-score calculation for sample “S”:

$$Z = \frac{(\text{Taxon rPM in } S) - (\text{Taxon mean rPM in Negative Controls})}{\text{Standard deviation for Taxon rPM in Negative Controls}}$$

# Additional controls

- Controls can also be used to create a background model to identify pathogens of interest (for example, cases vs controls)
- Sample controls can also be used to identify contaminants from sampling, but can be difficult to come by
- Background models can be applied to the sample report and the heatmap

Let's make a background model!

# Aggregate Score (Score)

- The Z-score is used to calculate an aggregate score (Score). The aggregate score is an empirical heuristic for ranking microbial matches by combining relative abundance and Z-score information at the species and genus levels. The aggregate score is calculated as follows:

$$\text{Score} = \frac{[\text{abs}(\text{genus Z for NT}) \times (\text{species Z for NT}) \times (\text{species rPM aligned in NT})]}{[\text{abs}(\text{genus Z for NR}) \times (\text{species Z for NR}) \times (\text{species rPM aligned in NR})]}$$

- If a taxon has a high aggregate score it indicates that a high abundance of reads matched to that taxon in the NT and/or NR database, while being less prevalent in the selected background model.